ECAG-Bench: A Benchmark for Constructive Algebraic Geometry Reasoning in Large Language Models

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Abstract

We introduce ECAG-Bench, a benchmark for evaluating the ability of large language models to perform *constructive reasoning* in algebraic geometry. Unlike existing mathematical benchmarks that focus on theorem proving or competition problems, ECAG-Bench requires models to construct concrete mathematical objects—schemes, sheaves, morphisms, and cohomology classes—satisfying specified properties. The benchmark comprises 332 problems across five chapters of algebraic geometry, spanning three difficulty levels, with a four-layer verification pipeline that validates properties of constructed objects rather than checking answer equivalence. We establish baselines using frontier language models and analyze their failure modes, finding that constructive algebraic geometry reasoning remains a significant challenge even for the most capable models.

Keywords

Large language models, mathematical reasoning, algebraic geometry, benchmark, constructive mathematics

1 Introduction

Mathematical reasoning has emerged as a central testbed for evaluating the capabilities of large language models (LLMs). Existing benchmarks primarily assess two modes of mathematical reasoning: *competition-style problem solving* [Hendrycks et al., 2021, Zhong et al., 2023] and *formal theorem proving* [Tsoukalas et al., 2024]. Both modes share a common structure: given a precisely stated problem, produce a proof or a unique numerical answer.

We identify a third mode—*constructive reasoning*—that is pervasive in mathematical practice yet largely absent from current benchmarks. Constructive reasoning asks: given a set of desired properties, can you *build* a concrete mathematical object that satisfies them? For example:

- Construct a smooth projective curve of genus 2 whose Jacobian has complex multiplication.
- Give an example of a flat morphism that is not smooth.
- Find a coherent sheaf on $\mathbf{P}^2$ with prescribed Hilbert polynomial.

This mode of reasoning is fundamentally different from verification. In verification, one checks whether a given object has a property; in construction, one must *synthesize* an object from scratch. The answer space is typically infinite and non-unique: many different objects may satisfy the given constraints, and the "difficulty" lies not in finding *the* answer but in finding *any* valid answer.

Algebraic geometry provides an ideal domain for testing constructive reasoning. The field is characterized by a rich interplay between abstract theory and concrete computation: theorems about schemes, sheaves, and cohomology become meaningful only when instantiated on specific geometric objects. Standard references [Hartshorne, 1977, Vakil, 2017, Stacks project authors, 2024] develop the theory through carefully chosen examples that illustrate both the power and the limitations of general results.

We introduce ECAG-Bench (**E**xamples and **C**ounterexamples in **A**lgebraic **G**eometry **Bench**mark), a dataset of 332 constructive problems in algebraic geometry. Each problem asks the model to construct a specific mathematical object—a scheme, a morphism, a sheaf, a cohomology class, or a numerical invariant—satisfying stated properties. The problems are drawn from a curated collection of examples and counterexamples that has been developed over several years as supplementary material for learners entering algebraic geometry.

Our key contributions are:

1. **A new benchmark** of 332 constructive algebraic geometry problems at three difficulty levels, covering five major areas of the subject.
2. **A four-layer verification pipeline** that validates properties of constructed objects rather than checking answer equivalence, using computer algebra systems (CAS) as the primary verification mechanism.
3. **Baseline evaluations** of frontier LLMs, revealing that constructive algebraic geometry reasoning remains a significant challenge even for the most capable models.
4. **A taxonomy of failure modes** specific to constructive mathematical reasoning, including fabricated objects, property confusion, and computational errors in invariant calculations.

2 Related Work

Mathematical reasoning benchmarks. The MATH dataset [Hendrycks et al., 2021] evaluates competition-level mathematics across seven subjects, with problems requiring numerical or symbolic answers. AGIEval [Zhong et al., 2023] includes math problems from standardized tests. These benchmarks test *problem-solving* ability but focus on high-school and undergraduate competition mathematics, where problems have unique answers and standard solution techniques.

Research-level mathematics. FrontierMath [Glazer et al., 2024] is the closest benchmark in spirit, comprising research-level mathematics problems with custom verification scripts. However, FrontierMath spans all areas of mathematics and does not specifically target constructive reasoning; many problems still ask for unique numerical answers. GHOSTS [Zheng et al., 2023] evaluates graduate-level science tasks using rubric-based scoring, but covers mathematics only as one component of a broader scientific evaluation.

Formal theorem proving. PutnamBench [Tsoukalas et al., 2024] formalizes Putnam competition problems in Lean, Isabelle, and Coq, testing whether LLMs can produce machine-verifiable proofs. AMBER [Liu et al., 2025] combines construction and verification in Lean 4, requiring models to both state and prove lemmas. These benchmarks operate in the formal verification paradigm, where correctness is checked by a proof assistant. ECAG-Bench differs by operating in the *informal* mathematical setting, using CAS verification of object properties rather than formal proof checking.

Interactive and multi-turn evaluation. Collins et al. [2024] propose evaluating mathematical capabilities through multi-turn interactions. MathArena [Balunović et al., 2025] uses recent competition problems with dual grading (automatic and human) to minimize contamination. ECAG-Bench currently uses single-turn evaluation but is designed to support multi-turn interaction in future work.

Constructive vs. verificational reasoning. The distinction between constructive and verificational reasoning is well-established in mathematics [Hartshorne, 1977] but has received limited

attention in the LLM evaluation literature. To our knowledge, ECAG-Bench is the first benchmark to systematically evaluate constructive reasoning in a specific mathematical domain, with property-based verification as the primary evaluation mechanism.

3 Dataset

3.1 Source Material

ECAG-Bench is derived from *Examples and Counterexamples in Elementary Algebraic Geometry*, a collection of worked examples originally written as supplementary material for learners entering algebraic geometry. The collection has been developed over several years and covers topics ranging from basic scheme theory to derived categories and moduli spaces.

The original material was written in LaTeX and subsequently converted to Markdown for web publication. Each example was then enhanced to meet a uniform quality standard: a self-contained mathematical narrative with complete constructions, precise references, and explicit computational details.

3.2 Content Enhancement

The content enhancement process involved restructuring each example into a natural mathematical exposition following a consistent quality standard:

1. **Narrative form:** Each example reads as a self-contained mathematical essay rather than a template with bullet points.
2. **Complete constructions:** All computational steps are shown explicitly, allowing independent verification.
3. **Benchmark problem extraction:** Each example is annotated with a structured benchmark problem specification, including the problem statement, expected answer type, difficulty level, and verification strategy.

All 332 entries have been enhanced to this standard, with zero stubs remaining.

3.3 Statistics

3.4 Difficulty Levels

Problems are assigned one of three difficulty levels:

Level 1 (Routine): The construction follows directly from definitions. A student who has read the relevant section of a standard textbook should be able to produce an answer.

Level 2 (Non-trivial): The construction requires applying a significant theorem or combining multiple standard results. This is the level of a good homework problem in a graduate course.

Level 3 (Deep): The construction requires either deep structural insight, the combination of results from different areas, or the knowledge of a specific "trick" that is not obvious from the problem statement.

3.5 Quality Control

Each entry has been reviewed for:

- Mathematical correctness of the stated construction
- Completeness of computational details
- Accuracy of the benchmark problem formulation
- Consistency of difficulty rating across the dataset
- Validity of the assigned verification strategy

Table 1: ECAG-Bench dataset statistics.

Metric	**Count**
Total entries	332
Substantive (with solution)	332
Stubs	0
By Chapter	
Basic Concepts	162
Computations	66
Moduli & Deformation	58
Cohomology	46
By Difficulty	
Level 1 (Routine)	77
Level 2 (Non-trivial theorem)	237
Level 3 (Deep/combining results)	18
By Question Type	
Example	243
Counterexample	66
Computation	14
Proof	9
By Verification Strategy	
CAS	309
Exact match	14
Rubric	9

4 Benchmark Design

4.1 Problem Types

ECAG-Bench problems fall into four categories:

Example: Construct a mathematical object with specified properties. ("Give an example of a flat morphism that is not smooth.")

Counterexample: Construct an object that violates a plausible-sounding statement. ("Give a counterexample to: every coherent sheaf on a smooth variety is locally free.")

Computation: Compute a specific invariant of a given object. ("Compute the Picard group of $\operatorname{Spec} \mathbb{Z}[\sqrt{-5}]$.")

Proof: Prove a statement about a specific object. ("Show that the Horrocks–Mumford bundle is stable.")

4.2 Four-Layer Verification Pipeline

Since constructions in algebraic geometry are non-unique, standard answer-matching is insufficient. ECAG-Bench employs a four-layer verification pipeline:

Layer 1: Parsing. The model's output is parsed to extract the claimed mathematical object. This layer uses regex and SymPy to determine whether the output is a well-formed mathematical expression (a polynomial, a variety defined by equations, a morphism given by coordinate functions, etc.).

Layer 2: CAS Verification. The extracted object is verified using computer algebra systems—primarily SageMath, with Singular and Macaulay2 for specialized computations. The CAS checks whether the constructed object actually satisfies the required properties: genus, smoothness, dimension, cohomology vanishing, etc. This is the *core layer*: 309 of 332 problems (93.1%) use CAS verification.

Layer 3: LLM-Assisted Judgment. For problems where CAS verification is insufficient (e.g., those requiring assessment of reasoning quality or non-triviality), a rubric-based LLM judge evaluates:

- (C1) Correctness: Is the constructed object valid?
- (C2) Completeness: Are all required properties verified?
- (C3) Non-triviality: Is the answer non-degenerate?
- (C4) Reasoning: Is the justification sound?

Layer 4: Human Expert Sampling. A random subset of model responses is reviewed by a human expert in algebraic geometry to calibrate the automated layers and catch systematic errors.

4.3 Verification Strategies

Each problem is assigned one of three verification strategies:

Strategy A (Exact): Problems with unique numeric or symbolic answers (e.g., "Compute $\dim H^1(X, \mathcal{O}_X)$"). Verified by exact match. 14 problems (4.2%).

Strategy B (CAS): The model outputs a mathematical object; CAS verifies its properties deterministically. This is the primary strategy. 309 problems (93.1%).

Strategy C (Rubric): Structured scoring on correctness, completeness, non-triviality, and reasoning validity. Used for proofs and complex constructions. 9 problems (2.7%).

4.4 Evaluation Protocol

Models are evaluated in a zero-shot, single-turn setting. Each problem is presented as a standalone prompt with the following format:

```
You are an expert in algebraic geometry. [Problem statement]. Provide
a concrete construction with full justification.
```

Responses are scored as:

- **Correct (1.0):** The constructed object is valid and satisfies all stated properties.
- **Partial (0.5):** The object is valid but missing some properties, or the justification is incomplete.
- **Incorrect (0.0):** The object is invalid, does not exist, or does not satisfy the required properties.

5 Experiments

[TODO: Run baseline experiments and fill in results.]

5.1 Models

We evaluate the following frontier models:

- Claude Opus 4 [Anthropic, 2024]
- GPT-4o [OpenAI, 2024]

- Gemini 2.5 Pro [Google DeepMind, 2024]

All models are evaluated via their respective APIs with default parameters (temperature 0, no system prompt modifications beyond the evaluation template).

5.2 Results

Table 2: Overall accuracy on ECAG-Bench by model and difficulty level. **[TODO: Fill with actual results.]**

Model	Overall	Level 1	Level 2	Level 3	Partial
Claude Opus 4	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GPT-4o	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Gemini 2.5 Pro	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 3: Accuracy by chapter. **[TODO: Fill with actual results.]**

Model	Basic	Cohomology	Computations	Moduli
Claude Opus 4	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GPT-4o	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Gemini 2.5 Pro	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 4: Accuracy by question type. **[TODO: Fill with actual results.]**

Model	Example	Counter-ex.	Computation	Proof
Claude Opus 4	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GPT-4o	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Gemini 2.5 Pro	[TODO:]	[TODO:]	[TODO:]	[TODO:]

6 Analysis

[TODO: Fill in after running experiments.]

6.1 Failure Mode Taxonomy

Based on preliminary evaluation, we identify the following failure modes specific to constructive algebraic geometry reasoning:

F1: Fabricated objects. The model produces a mathematical object that does not exist or is not well-defined. For example, claiming that a certain polynomial defines a smooth curve when it is in fact singular, or referencing a "well-known" variety that does not appear in the literature.

F2: Property confusion. The model constructs a valid object but confuses the required properties. For example, producing a smooth but projective variety when asked for a smooth non-projective variety, or computing the Picard group when asked for the class group.

F3: Computational errors. The model follows a correct strategy but makes arithmetic or algebraic errors in the computation. This is particularly common in problems involving intersection numbers, Chern class computations, and cohomology calculations.

F4: Trivial or degenerate answers. The model produces a technically correct but trivially degenerate answer that fails to demonstrate understanding. For example, answering "give an example of a flat morphism" with the identity morphism.

F5: Hallucinated theorems. The model invokes a theorem that does not exist, or misattributes a result to a standard reference. This is distinct from F1 in that the constructed object may be valid, but the justification relies on a non-existent result.

F6: Incomplete verification. The model constructs a plausible object and provides some justification, but fails to verify all the required properties. For example, showing that a morphism is flat without checking that it is not smooth.

6.2 Difficulty Calibration

[TODO: Analyze whether model performance correlates with assigned difficulty levels. Check for problems where difficulty assignment may need revision.]

6.3 Chapter-Level Analysis

[TODO: Analyze which areas of algebraic geometry are most challenging for LLMs. Hypothesize reasons based on training data availability and problem structure.]

7 Conclusion

We have introduced ECAG-Bench, a benchmark of 332 constructive problems in algebraic geometry designed to evaluate a mode of mathematical reasoning—object construction—that is central to mathematical practice but underrepresented in existing LLM benchmarks. Our four-layer verification pipeline, anchored by CAS-based property verification, enables reliable evaluation of non-unique mathematical constructions.

Limitations. ECAG-Bench focuses on algebraic geometry, which limits the generalizability of our findings to other areas of mathematics. The CAS verification layer, while powerful, cannot verify all properties—some require human judgment or more sophisticated formal methods. Our current evaluation uses single-turn prompting; multi-turn interaction may significantly improve model performance.

Future work. We plan to:

1. Expand the benchmark to cover additional areas (arithmetic geometry, algebraic topology, commutative algebra).
2. Implement the full CAS verification pipeline with automated SageMath/Macaulay2 scripts.
3. Evaluate models in multi-turn interactive settings, where they can refine constructions based on CAS feedback.
4. Investigate whether fine-tuning on algebraic geometry textbooks improves constructive reasoning.
5. Release a public leaderboard for tracking progress on ECAG-Bench.

Broader impact. Constructive mathematical reasoning is a prerequisite for using LLMs as research tools in mathematics. If models cannot construct examples and counterexamples, they cannot assist mathematicians in the exploratory phase of research—the phase where intuition is built and conjectures are tested. ECAG-Bench provides a concrete measure of progress toward this goal.

References

Anthropic. The Claude 3 model family: Opus, Sonnet, Haiku. 2024. Model card available at `https://www.anthropic.com`.

Mislav Balunović, Nikola Jovanović, Robin Staab, Mark Vero, and Martin Vechev. MathArena: Evaluating llms on uncontaminated math competitions. 2025. ETH Zurich, `https://matharena.ai/`.

Katherine M Collins, Albert Q Jiang, Simon Frieder, Lionel Wong, Miri Zilka, Umang Bhatt, Thomas Lukasiewicz, Yuhuai Wu, Joshua B Tenenbaum, William Hart, et al. Evaluating language models for mathematics through interactions. *arXiv preprint arXiv:2306.01694*, 2024.

Elliot Glazer, Ege Erdil, Tamay Besiroglu, Diego Chicharro, Evan Chen, Alex Gunning, Caroline Falkman Olsson, Jean-Stanislas Denain, Anson Ho, Emily de Oliveira Santos, et al. Frontiermath: A benchmark for evaluating advanced mathematical reasoning in ai. *arXiv preprint arXiv:2411.04872*, 2024.

Google DeepMind. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2024.

Robin Hartshorne. *Algebraic Geometry*. Springer, 1977.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *NeurIPS*, 2021.

Yiming Liu et al. AMBER: Construction and verification benchmark for mathematical reasoning in Lean 4. *arXiv preprint arXiv:2602.01291*, 2025.

OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2024.

The Stacks project authors. The stacks project. `https://stacks.math.columbia.edu`, 2024.

George Tsoukalas, Jasper Lee, John Jennings, Jimmy Xin, Michelle Ding, Michael Jennings, Amitayush Thakur, and Swarat Chaudhuri. PutnamBench: Evaluating neural theorem-provers on the putnam mathematical competition. *arXiv preprint arXiv:2407.11214*, 2024.

Ravi Vakil. *The Rising Sea: Foundations of Algebraic Geometry*. 2017. Available at `https://math.stanford.edu/~vakil/216blog/`.

Yifan Zheng et al. GHOSTS: Graduate-level holistic open-ended science tasks. In *NeurIPS 2023 Datasets and Benchmarks Track*, 2023.

Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yaobo Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. AGIEval: A human-centric benchmark for evaluating foundation models. *arXiv preprint arXiv:2304.06364*, 2023.